

AI503

Machine Learning

Lecture 3: Supervised Learning 1 -SVM

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Support Vector Machine (SVM) Algorithm

- Support Vector Machine (SVM) is a powerful machine learning algorithm used for linear or nonlinear classification, regression, and even outlier detection tasks.
- SVMs can be used for a variety of tasks, such as text classification, image classification, spam detection, handwriting identification, gene expression analysis, face detection, and anomaly detection.

Support Vector Machine (SVM) Algorithm

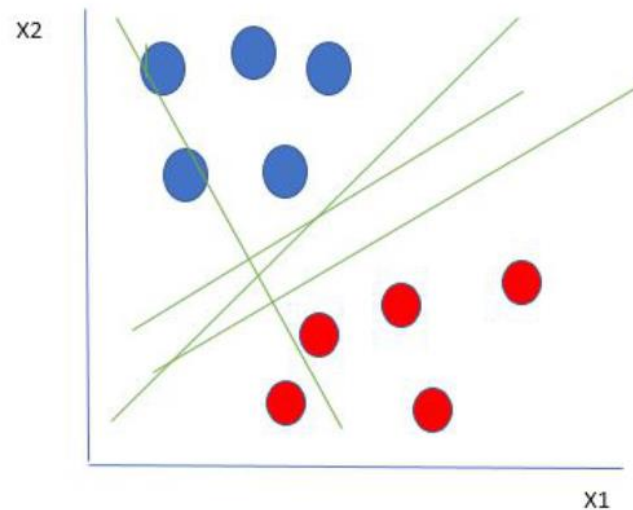
- SVMs are adaptable and efficient in a variety of applications because they can manage high-dimensional data and nonlinear relationships.
- SVM algorithms are very effective as we try to find the maximum separating hyperplane between the different classes available in the target feature.

Support Vector Machine

- Support Vector Machine (SVM) is a supervised machine learning algorithm used for both classification and regression.
- The main objective of the SVM algorithm is to find the optimal hyperplane in an N-dimensional space that can separate the data points in different classes in the feature space.

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- The hyperplane tries that the margin between the closest points of different classes should be as maximum as possible.
 - The dimension of the hyperplane depends upon the number of features.
 - If the number of input features is two, then the hyperplane is just a line.
 - If the number of input features is three, then the hyperplane becomes a 2-D plane. It becomes difficult to imagine when the number of features exceeds three.

- Let's consider two independent variables x_1 , x_2 , and one dependent variable which is either a blue circle or a red circle.



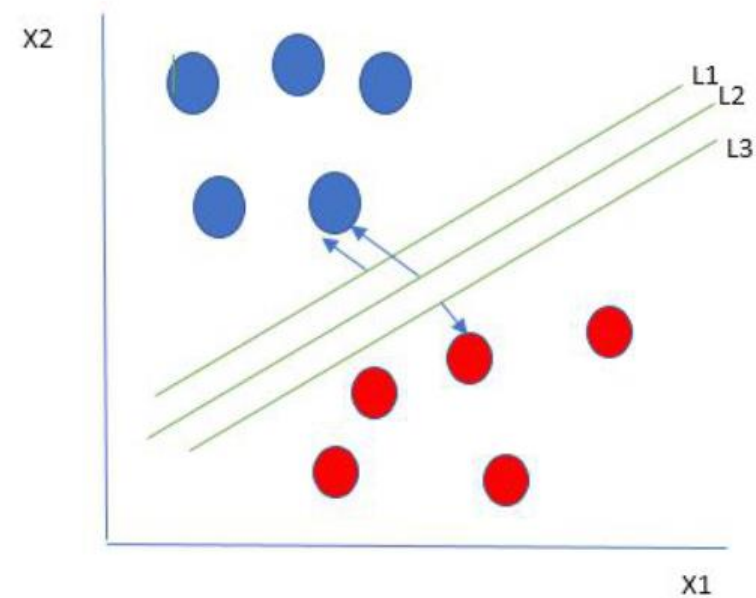
Linearly Separable Data points

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- From the figure in previous slide it's very clear that there are multiple lines (our hyperplane here is a line because we are considering only two input features x_1 , x_2) that segregate our data points or do a classification between red and blue circles.
 - So how do we choose the best line or in general the best hyperplane that segregates our data points?



How does SVM work?

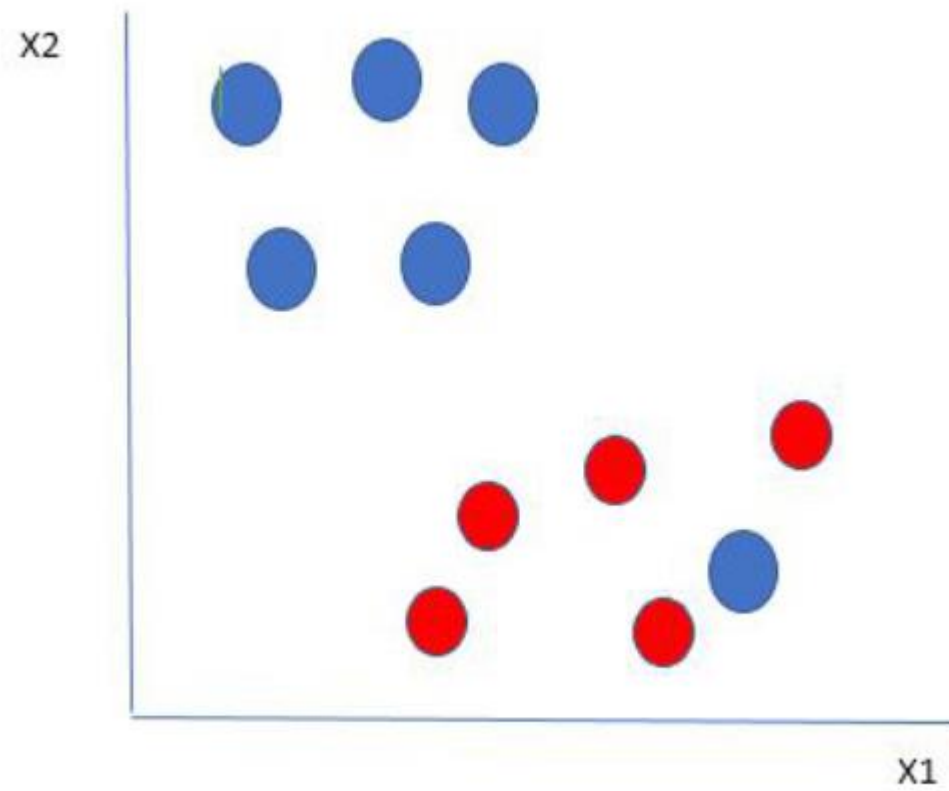
- One reasonable choice as the best hyperplane is the one that represents the largest separation or margin between the two classes.



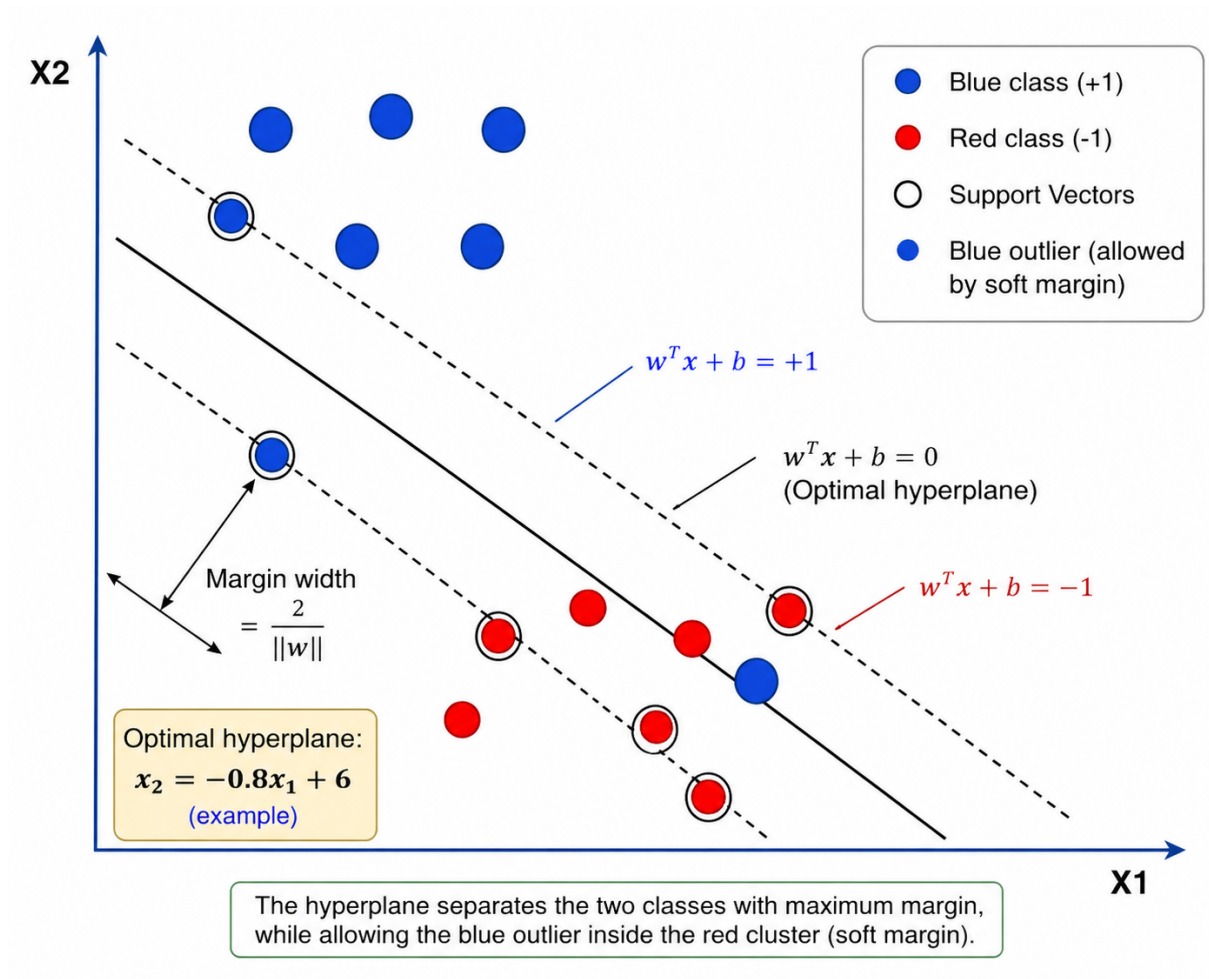
Multiple hyperplanes separate the data from two classes

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- So we choose the hyperplane whose distance from it to the nearest data point on each side is maximized. If such a hyperplane exists it is known as the **maximum-margin hyperplane/hard margin**. So from the above figure, we choose L2. Let's consider a scenario like shown below





Selecting hyperplane for data with outlier



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- So in this type of data point what SVM does is, finds the maximum margin as done with previous data sets along with that it adds a penalty each time a point crosses the margin. So the margins in these types of cases are called **soft margins**. When there is a soft margin to the data set, the SVM tries to minimize $(1/\text{margin} + \lambda(\sum \text{penalty}))$. Hinge loss is a commonly used penalty. If no violations no hinge loss. If violations hinge loss proportional to the distance of violation.

Support Vector Machine Terminology

1. **Hyperplane:** Hyperplane is the decision boundary that is used to separate the data points of different classes in a feature space. In the case of linear classifications, it will be a linear equation i.e. $wx+b = 0$.
2. **Support Vectors:** Support vectors are the closest data points to the hyperplane, which makes a critical role in deciding the hyperplane and margin.
3. **Margin:** Margin is the distance between the support vector and hyperplane. The main objective of the support vector machine algorithm is to maximize the margin. The wider margin indicates better classification performance.

Support Vector Machine Terminology

1. **Kernel:** Kernel is the mathematical function, which is used in SVM to map the original input data points into high-dimensional feature spaces, so, that the hyperplane can be easily found out even if the data points are not linearly separable in the original input space. Some of the common kernel functions are linear, polynomial, radial basis function(RBF), and sigmoid.
2. **Hard Margin:** The maximum-margin hyperplane or the hard margin hyperplane is a hyperplane that properly separates the data points of different categories without any misclassifications.
3. **Soft Margin:** When the data is not perfectly separable or contains outliers, SVM permits a soft margin technique. Each data point has a slack variable introduced by the soft-margin SVM formulation, which softens the strict margin requirement and permits certain misclassifications or violations. It discovers a compromise between increasing the margin and reducing violations.

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- Consider a binary classification problem with two classes, labeled as +1 and -1. We have a training dataset consisting of input feature vectors X and their corresponding class labels Y .
 - The equation for the linear hyperplane can be written as:

$$w^T x + b = 0$$

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- The vector $\mathbf{W}$ represents the normal vector to the hyperplane. i.e the direction perpendicular to the hyperplane. The parameter $\mathbf{b}$ in the equation represents the offset or distance of the hyperplane from the origin along the normal vector $\mathbf{w}$.

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- The distance between a data point x_i and the decision boundary can be calculated as:

$$d_i = \frac{w^T x_i + b}{\|w\|}$$

- where $\|w\|$ represents the Euclidean norm of the weight vector w . Euclidean norm of the normal vector W

- For Linear SVM classifier :

$$\hat{y} = \begin{cases} 1 & : w^T x + b \geq 0 \\ 0 & : w^T x + b < 0 \end{cases}$$

Types of Support Vector Machine

- Based on the nature of the decision boundary, Support Vector Machines (SVM) can be divided into two main parts:
- **Linear SVM:** Linear SVMs use a linear decision boundary to separate the data points of different classes. When the data can be precisely linearly separated, linear SVMs are very suitable. This means that a single straight line (in 2D) or a hyperplane (in higher dimensions) can entirely divide the data points into their respective classes. A hyperplane that maximizes the margin between the classes is the decision boundary.

Types of Support Vector Machine

- **Non-Linear SVM:** Non-Linear SVM can be used to classify data when it cannot be separated into two classes by a straight line (in the case of 2D). By using kernel functions, nonlinear SVMs can handle nonlinearly separable data. The original input data is transformed by these kernel functions into a higher-dimensional feature space, where the data points can be linearly separated. A linear SVM is used to locate a nonlinear decision boundary in this modified space.

Popular kernel functions in SVM

- The SVM kernel is a function that takes low-dimensional input space and transforms it into higher-dimensional space, ie it converts nonseparable problems to separable problems. It is mostly useful in non-linear separation problems. Simply put the kernel, does some extremely complex data transformations and then finds out the process to separate the data based on the labels or outputs defined.

Popular kernel functions in SVM

Linear : $K(w, b) = w^T x + b$

Polynomial : $K(w, x) = (\gamma w^T x + b)^N$

Gaussian RBF: $K(w, x) = \exp(-\gamma \|x_i - x_j\|^n)$

Sigmoid : $K(x_i, x_j) = \tanh(\alpha x_i^T x_j + b)$

In practice,

- There are some generic kernel functions

Gaussian RBF	$k(\mathbf{x}, \mathbf{y}) = \exp\left(\frac{-\ \mathbf{x} - \mathbf{y}\ ^2}{\gamma}\right)$
Polynomial	$((\mathbf{x} \cdot \mathbf{y}) + \theta)^d$
Sigmoidal	$\tanh(\kappa(\mathbf{x} \cdot \mathbf{y}) + \theta)$
inv. multiquadric	$\frac{1}{\sqrt{\ \mathbf{x} - \mathbf{y}\ ^2 + c^2}}$

- Customized kernels may achieve better performance for particular problems, but in general the above will do.

4. Weighting Methods.

4.1 TF-IDF Technique.

TF-IDF works by determining the relative frequency of words in a specific document compared to the inverse proportion of that word over the entire document corpus. Intuitively, this calculation determines how relevant a given word is in a particular document. Words that are common in a single or a small group of documents tend to have higher TF-IDF numbers than common words such as articles and prepositions. [Ram02]

TF : stands for term frequency and simply means that each term component in a term vector is multiplied by the frequency by which that term occurs in the document.

IDF: stands for inverse document frequency which is defined as:

$$\log_2 \left(\frac{N}{n_j} \right)$$

Where:

N : Total number of document under study.

n_j : Number of documents containing term j

And the TF-IDF is simply the multiplication of TF by the IDF:

$$TF_IDF = TF * \log_2 \left(\frac{N}{n_j} \right)$$

Using the IDF weight favor terms that occur in relatively few documents, so it has discriminative power over TF and the use of the logarithm of IDF rather than IDF directly is motivated by the desire to make the weight relatively insensitive to the exact number of document N . [Han01]

Advantages of SVM

- Effective in high-dimensional cases.
- Its memory is efficient as it uses a subset of training points in the decision function called support vectors.
- Different kernel functions can be specified for the decision functions and its possible to specify custom kernels.

VM implementation in Python

- Predict if cancer is Benign or malignant. Using historical data about patients diagnosed with cancer enables doctors to differentiate malignant cases and benign ones are given independent attributes.

VM implementation in Python

- **Steps**
 - Load the breast cancer dataset from `sklearn.datasets`
 - Separate input features and target variables.
 - Build and train the SVM classifiers using RBF kernel.
 - Plot the scatter plot of the input features.
 - Plot the decision boundary.
 - Plot the decision boundary